Evaluation and Feedback Engine

Design Rationale and Proposed Pipeline

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1. Problem Statement

Given a question, an answer written by a human, and a reference answer produced by a large language model, build a system that evaluates the human answer along interpretable dimensions rather than by surface similarity to the reference, weights those dimensions according to what the question actually rewards, and returns actionable feedback rather than only a score.

2. Design Challenges and Resolutions

The following questions were raised while working out the design. Each is answered briefly, with the supporting literature named where the answer draws on it.

1. What does the system compare the answer against, when the questions are not pre-defined?

Not against the LLM answer. The system compares against a rubric that is generated for that specific question at the time the question is generated. The LLM answer then becomes one instance of a rubric-satisfying answer rather than the target.

This matters because a human answer can be correct by a route the LLM never took, or better than the LLM answer. Rubric comparison handles both cases; answer-to-answer comparison marks them wrong.

Source: Pathak et al., Rubric Is All You Need (ICER 2025), which shows that question-specific rubrics outperform question-agnostic rubrics, most clearly on problems admitting multiple valid approaches.

2. What is the user input, and does the topic the user picks decide how the answer is graded?

The user picks a topic only, for example history, technology or ethics. The topic makes the question interesting to the user but tells us very little about how to grade it.

What decides the grading is the task type: factual, analytical, explanatory or creative. Topic and task type are orthogonal. A history topic can yield a pure factual question or a pure argumentative one. The task type is therefore declared by the generator at generation time, as part of the same structured output that produces the question and rubric, rather than inferred later.

This also gives a free consistency check. If the generator declares an analytical task but emits a rubric made entirely of factual checkboxes, something has gone wrong.

3. Does the category matter for evaluation, and in what way?

Yes, but it changes the weights of the dimensions, not which dimensions exist. It also changes the shape of the rubric.

Factual tasks are dominated by correctness and use checklist rubrics. Analytical tasks are dominated by reasoning and use claim-and-evidence rubrics. Explanatory tasks sit between the two. Creative tasks are dominated by originality and expression, cannot use checklists at all, and are the one case where divergence from the LLM answer is a positive signal rather than an error.

Four task types is the right number for a project of this size. Fewer forces creative work into a category where it does not belong; more splits distinctions that cannot be validated.

4. Which dimensions is the answer scored on?

Four: correctness, completeness, reasoning and expression. These stay constant across categories, and only their weights change.

5. Should the rubric be applied as a whole, or one criterion at a time?

As a whole, in a single call, with the per-criterion detail extracted from the structured output of that same call.

The intuitive choice is the opposite. Judging one criterion at a time sounds more stable and more granular. The evidence says it buys that stability at the cost of systematic harshness: an evaluator shown a single criterion in isolation applies

stricter standards and tends to award zero unless the answer fully satisfies that criterion. Seeing the whole rubric gives the evaluator enough context to infer intent.

For a system whose purpose is feedback to learners, an unjustly harsh grader is worse than a slightly noisy one.

Source: Pathak et al. (ICER 2025). Their Pointwise Rubric Evaluation recorded a leniency of -0.329 against human graders, roughly 11.5 marks lower on a 35 mark scale, while Complete Rubric Evaluation, fed the entire rubric at once, recorded 0.082. Their stated conclusion is that the evaluator should be given the entire rubric as evaluation context.

6. How do we handle an answer that is correct but takes a route the rubric did not list?

The rubric lists several valid solution paths rather than one. Before scoring, the system identifies which path the answer took, with a confidence score, and then grades against that path's criteria.

This directly addresses the case where the human answered correctly but differently from the LLM.

Source: Pathak et al. (ICER 2025) use exactly this mechanism on their algorithms dataset, where rubrics enumerate multiple solutions and an approach-identification step with a confidence value precedes grading.

7. What happens when the human answer is better than the rubric anticipated?

This is the rubric ceiling problem, and it has no settled solution in the literature. Multi-path rubrics raise the ceiling but do not remove it, since they only cover approaches that were anticipated.

Our approach is a two-stage amendment mechanism. When path identification returns low confidence across all listed paths, a separate call is made that sees only the question and the answer, and is asked a single narrow question: does this answer contain a valid approach the rubric does not cover? It must return the proposed criteria and a justification. It is deliberately not shown the existing scores, so it is not anchored on the conclusion that a mismatch means an error.

It is worth noting that some evaluation guidance takes the opposite position and treats the rubric ceiling as deliberate, instructing judges not to reward correct outside knowledge, precisely because judges tend to over-reward plausible-sounding additions. Our design accepts the risk but contains it through verification.

Source: Xie et al., Grade Like a Human (2024), is the closest existing mechanism. Its rubrics are built considering both the question and the student answers rather than the question alone, and it adds a post-grading review stage for fairness.

8. How do we stop the rubric from bending towards whatever the user happened to write?

This is the main risk of combining a frozen multi-path rubric with answer-conditioned rubric generation. Done naively, a confidently wrong answer receives its own custom path and scores full marks. The system becomes a flattery machine.

The design resolves this with an asymmetry. The pre-written rubric is binding. The answer-conditioned stage can only propose an amendment, and every amendment must pass a verification step before it is used. Verification is performed by a different model under adversarial framing, instructed to try to break the proposed path. If it does not survive, the answer is graded against the original rubric and the divergence is reported as feedback rather than passed without penalty.

One difference from the prior work should be noted. Grade Like a Human adapts its rubrics using a whole cohort of answers. Our system sees one answer at a time with no cohort, which is a harder problem, and is why verification matters more here than it did there.

9. How are we going to evaluate a creative answer?

Not by asking the system to score creativity. The rubric for a creative prompt becomes constraint satisfaction plus craft checks, expressed as many binary, observable tests: were the brief's constraints met, is there development or a turn, does it show rather than tell, does it avoid cliche. Only the aggregate is reported, never the individual test results.

For originality specifically, the one defensible signal available is a negative test. At generation time, before any answer is seen, the generator is asked to produce the five most obvious ways to answer the prompt. Overlap with that list is evidence of unoriginality. Detecting cliche is far more reliable than positively detecting originality.

Path identification and amendment verification are switched off entirely for creative tasks. There are no paths to identify and nothing for the verifier to verify against. This should be stated in the report rather than allowing the machinery to run and produce noise.

Creative evaluation will be the weakest part of the system, and that should be reported honestly rather than concealed.

Source: Chakrabarty et al., Torrance Tests of Creative Writing, operationalise creativity through fourteen binary tests across fluency, flexibility, originality and elaboration; per-test agreement between raters is only moderate while aggregate agreement is strong, which is

what licenses reporting the set rather than the parts. Rethinking Creativity Evaluation (2025) reports that models struggle to assess conceptual creativity when prompted directly and are biased towards one-sided predictions, labelling most content non-original at a reasonable threshold. Chhun et al. (EMNLP Findings 2023) report only moderate agreement between human and model ratings on a creative writing rubric.

10. How do we stop the judge from favouring answers that resemble its own writing?

Models rate text that resembles their own output more highly. Since one model generates the question, the rubric and the reference answer, a human answer phrased differently would be penalised for style rather than substance.

The judge model is therefore drawn from a different family than the generator, or a small ensemble is used. Related to this, all criteria are written as binary and evidence-anchored statements rather than open ratings, because loosely worded criteria produce scores that swing between runs.

Source: Pathak et al. (ICER 2025) implement ensembling across models with majority voting, and report that gains plateau at around three to four high-capability models.

11. How is the feedback given?

Prioritised, not exhaustive. Every criterion produces a verdict and a justification, and the temptation is to return all of it. Feedback that lists twelve problems fixes none of them. Two or three failures are selected, ranked by what cost the most, with a missed core criterion outranking a stylistic one.

Each point must be specific rather than evaluative, naming what in the answer caused the judgement. Each must be forward-looking, implying an action rather than only a verdict. The feedback must also say what worked, not for encouragement but for calibration, so that the learner does not rewrite the correct parts.

Feedback must reference the rubric criterion and never the LLM's prose. Phrasing feedback as a comparison with the reference answer reintroduces the self-preference problem through the back door.

12. Should the user see a score, and in what form?

Per-dimension coarse bands, no single aggregate, feedback placed above the scores.

A visible number suppresses engagement with the feedback, which is one of the more consistently replicated findings in assessment research. A single aggregate is also dishonest in our case, because the dimension weights are a design choice rather than a measured quantity, so collapsing four judgements into one number destroys information and adds false authority.

Bands are kept coarse, three or four levels per dimension rather than a percentage, because the system cannot reliably distinguish adjacent fine-grained values. The internal scoring stays continuous, since correlation and leniency need it; the coarsening is a presentation decision, not a measurement one.

Source: Rubric-Conditioned LLM Grading (2026) reports that off-the-shelf models are effective at binary grading but struggle with the nuance required for three-way and five-way partial credit.

13. How do we validate the system without a large human-graded dataset?

Leniency and correlation are both relative metrics and need base data. We do not have graders available at the scale the prior work used, so validation is assembled from three cheaper sources.

First, a borrowed corpus of short answers that already carries human scores. This loses the generated-question part of the pipeline but validates the engine itself. Second, synthetic perturbation: take a good answer and degrade it in controlled ways, dropping a key point, inserting a factual error, or damaging only the expression. The true score is unknown but the direction of change is known, so the test becomes whether the engine detects the injected flaw and attributes it to the correct dimension. Third, a small human-graded set, around forty to sixty answers across one or two categories, which is enough for a leniency figure with honest error bars.

Pairwise ranking is a cheaper alternative to absolute scoring if human time runs short, since deciding which of two answers is better takes seconds.

Both correlation and leniency must be reported. Correlation measures ordering and cannot see systematic offset; a grader that marks every answer ten points low still correlates almost perfectly.

Source: Leniency is introduced in Pathak et al. (ICER 2025) as the mean normalised signed error against expert scores, with a suggested target within plus or minus 0.1. Candidate corpora include ASAP-SAS, the SemEval-2013 SciEntsBank and Beetle sets, and RiceChem.

14. How do we know the amendment verifier actually works?

It is tested directly, with synthetic proposals whose validity is known in advance. The positive class consists of approaches that are genuinely valid but absent from the rubric, produced by asking a separate model to solve the question unlike the listed paths.

The negative classes are deliberately varied, because they fail differently: confident nonsense that is fluent and wrong; a correct conclusion reached by invalid reasoning; a restatement of an existing path that path identification failed to match; and a real approach containing a hole.

Accuracy is the wrong metric here, since the classes are imbalanced and the errors are not equally costly. A false accept, where nonsense receives its own custom path, is much worse than a false reject, where a good answer is graded strictly and told why. Precision and recall on the positive class are therefore reported separately, along with the rejection rate for each negative class.

Roughly twenty questions with five proposals each gives one hundred items, which is small enough to verify by hand. Running the same set under both adversarial and neutral verifier prompts isolates a single variable and produces a result that is interesting either way.

15. What is logged, and what is shown to the user about an amendment?

The full trace is logged for every evaluation: the hash of the frozen rubric, the path confidences, whether the amendment stage fired, the proposed criteria verbatim, the verifier verdict and its reasoning, which prompt framing was used, and which rubric the final score came from. This makes every score auditable, and the accumulated logs become a dataset in their own right, giving real frequency figures for how often the rubric ceiling is reached in each category.

Accepted amendments are written back to a per-question rubric store, so the same question does not re-derive the same path repeatedly. This is a cheap approximation of cohort-based rubric adaptation, accumulated over time instead of across a batch.

What the user sees is deliberately restrained. Announcing that the user found an approach the rubric did not anticipate is flattering, becomes gameable once noticed, and is actively harmful when the verifier is wrong. The system instead states factually that the approach differs from the standard routes and was evaluated on its own terms, then gives normal feedback. A rejected amendment is given a distinct voice: the answer diverged, was checked, and did not hold up, with the specific reason. That is far more useful than a generic judgement of incorrectness, and it is only possible because the amendment path was run at all.

3. Points Deliberately Left Open

Two decisions remain unresolved and should be settled before implementation.

- Whether the path identification and amendment stages run on every answer or only on demand, which is a cost question.
- Whether the LLM reference answer is shown to the user at all. In the pipeline as designed it performs no work in the evaluation path, which is worth noting given that it was the starting point of the original project idea.

4. Proposed Pipeline

Stage 1: Generation, performed without sight of any answer

1. The user selects a topic.
2. The generator produces, in a single structured output, the question, the declared task type, the rubric, and, for creative tasks only, the list of the five most obvious answers.
3. The rubric shape follows the task type: a checklist for factual, multi-path claim-and-evidence for analytical, multi-path for explanatory, and binary craft tests for creative.
4. The rubric is frozen and hashed.
5. The LLM writes its reference answer afterwards, against the frozen rubric.

Stage 2: Evaluation of a submitted human answer

6. Path identification returns a confidence value. This step is skipped for creative tasks.
7. If confidence is low across all paths, an amendment proposal is generated. It sees only the question and the answer, not the scores.
8. The proposal is verified by a different model under adversarial framing. If accepted, the amended rubric is used. If rejected, the original rubric is used and the divergence is reported as feedback.
9. Scoring is performed in a single whole-rubric call by a judge model from a different family than the generator, returning per-criterion verdicts and justifications as structured output.
10. Aggregation and the application of category weights happen outside the model, in code.

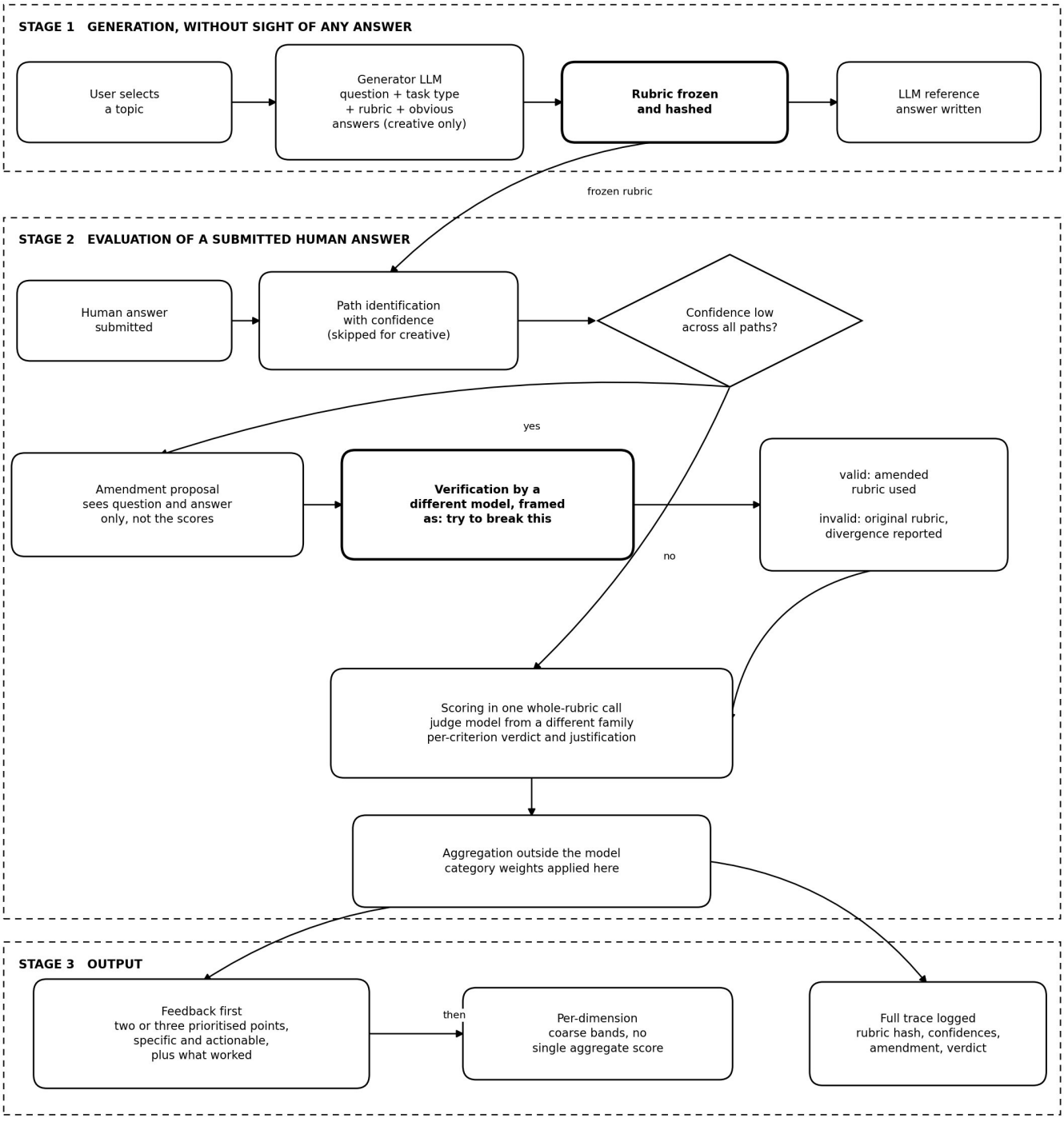
Stage 3: Output

11. Feedback is presented first: two or three prioritised points, each specific and actionable, together with what worked.
12. Per-dimension coarse bands follow. No single aggregate score is shown.
13. The full trace is logged.

Validation of the system

14. A borrowed graded corpus supplies correlation and leniency figures.
15. Synthetic perturbation tests flaw detection and the precision and recall of the verifier.
16. A small human-graded set supplies an independent leniency figure.

5. Pipeline Diagram



6. References

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